

Surya Batchu

Software Development Engineer

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SUMMARY

Software Development Engineer with 2+ years of experience designing and building large-scale, distributed, multi-tier, and highly available systems using Java and AWS. Strong in data structures, algorithms, and object-oriented design, with experience in multi-threaded, low-latency backend services. Skilled in end-to-end SDLC, including system design, coding, testing, code reviews, build processes, deployment, and operations. Focused on building scalable, resilient, and production-ready systems with strong emphasis on performance optimization, maintainability, and service reliability.

SKILLS

Languages: Java, Python, JavaScript (ES6+), TypeScript, Go, SQL, HTML5, CSS3.

Libraries & Frameworks: Spring Boot, Node.js, Express.js, React.

Backend & APIs: Distributed Systems, Microservices Architecture, RESTful APIs, System Design, Scalable Architecture, Concurrency, Multi-threading, Asynchronous Processing, Event-Driven Systems, Caching.

Database: PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, SQL Query Optimization, Data Modeling, Schema Design.

Cloud & DevOps: AWS (ECS, EC2, CloudWatch, Lambda, S3, RDS, SQS, Step Functions), Docker, CI/CD, Linux, Git

Tools & Testing: Unit Testing, JUnit, Mockito, Test Automation, API Testing.

Core Fundamentals: Data Structures, Algorithms, Object-Oriented Design, Design Patterns, Scalability, High Availability, Fault Tolerance, Performance Optimization, Thread Safety, Throughput Optimization, Latency Reduction, Code Maintainability

EXPERIENCE

Software Development Engineer (Graduate Assistant)

January 2024 – December 2025

Agricultural and Applied Economics department, Texas Tech University

- Developed scalable distributed backend services using Java and Spring Boot, building REST APIs for applications supporting 500+ users, ensuring high availability, low latency, and production-ready scalable architecture in production systems.
- Built multi-threaded and concurrent request-handling components, optimizing performance under high-load scenarios while ensuring thread safety, synchronization, and achieving latency reduction across distributed backend services.
- Implemented JWT-based authentication and role-based access control (RBAC), while designing optimized PostgreSQL schemas and queries, improving data consistency, system reliability, and service reliability in production applications.
- Wrote unit tests using JUnit and Mockito, improving code coverage and reducing defects.
- Participated in code reviews, Git-based source control management, and CI/CD build pipelines.
- Performed debugging, root cause analysis, and performance tuning to ensure production system reliability.

Junior Software Developer

July 2021 - September 2023

Growisto, India

- Developed large-scale distributed microservices using Spring Boot on AWS, supporting 10K+ users, designing systems with scalable architecture and resilient system design ensuring service reliability across distributed environments.
- Designed and implemented REST and GraphQL APIs, enabling seamless integration across distributed services while improving system scalability, performance, and reliability for customer-facing applications handling high concurrent traffic.
- Built asynchronous event-driven systems using AWS SQS and implemented multi-threaded processing, improving scalability, decoupling services, and optimizing throughput under high concurrent workloads in distributed backend systems.
- Improved system performance by ~40% through API optimization, caching strategies, and database query tuning.
- Implemented CloudWatch monitoring and alerting, enabling proactive detection and resolution of production issues.
- Contributed across full SDLC including coding, testing, build processes, deployment, and system operations.

PROJECTS

LifeLine - Smart Hospital Management System (Java · Spring Boot · JWT · MySQL)

- Designed 50+ RESTful APIs with JWT and RBAC, adding input validation, error handling, and OpenAPI docs to ensure secure, scalable operations across patient, staff, and admin workflows, enabling 500+ daily bookings with 99% uptime.
- Built a responsive multi-role portal with real-time scheduling, streamlining workflow and boosting patient satisfaction by 20%.

Zesty - A grocery selling app (Full-Stack MERN application)

- Developed a full-stack application using React, TypeScript, and Express, designing RESTful APIs and integrating Stripe payments and Gemini 2.5 REST API chatbot to deliver a scalable, consumer-facing platform with secure transaction handling.
- Optimized MongoDB queries and API performance to reduce response latency and ensure reliable user experiences.

EDUCATION

Master of Science in Computer Science, Texas Tech University, Texas - 3.9

January 2024 – December 2025

Bachelor of Technology in Electronics and Communication Engineering, NIT Durgapur, India

July 2017 - May 2021